Microbial Community Profiling with Differential Analysis

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Overview

This practical introduces students to taxonomic microbiome analysis using sequencing-derived abundance tables (for example from **Sylph**), focusing on: - Community structure - Diversity metrics - Differential abundance and prevalence

The workflow uses: - **phyloseq** for data handling and diversity analysis - **microViz** for transformation and ordination - **MaAsLin3** for statistical modelling

Questions addressed in this session

1. How does alpha diversity differ between children with severe malaria (SM) and Community Controls (CC)?
2. Do samples cluster by clinical group in beta diversity (ordination space)?
3. Which taxa dominate the microbial community at different taxonomic levels?
4. Are there community-wide shifts associated with disease status?
5. Which bacterial taxa are differentially abundant between SM and CC?
6. Do results highlight biologically relevant taxa (e.g. Enterobacteriaceae species)?

Expected learning points

- Taxonomic profiles describe who is present, not just function
- Alpha diversity reflects within-sample diversity
- Beta diversity reflects between-sample differences
- Ordination methods (MDS (Multidimensional Scaling)) reveal clustering patterns
- Differential prevalence and abundance requires careful statistical modelling
- Compositionality must be considered (e.g. Centered Log-Ratio (CLR) transformation)

Prerequisites

This module assumes:

- Basic familiarity with R / RStudio
- A Sylph-derived abundance table
- Sample metadata
- Required packages installed

If needed, install missing packages first: - install.packages("BiocManager") - BiocManager::install("phyloseq") - .libPaths("~/R/library")

```r
# ==========================================
# LIBRARIES
# ==========================================
library(readr)
library(tidyverse)
library(stringr)
library(tibble)
library(data.table)
library(phyloseq)
library(microbiome)
library(ggpubr)
library(microViz)
library(corncob)
library(ggthemes)
library(gghalves)
library(microbiomeutilities)
library(maaslin3)
```

```r
# ==========================================
# READ DATA
# ==========================================
abundance_data <- read.csv("/Users/LuicerAnneIngasia/Desktop/wcs/sylph_read_based.csv", check.names = F

metadata <- read.csv("/Users/LuicerAnneIngasia/Desktop/wcs/Bednarski_2025_cleaned_metadata_final.csv", 
  column_to_rownames("metagm_id")

# Inspect data
head(abundance_data[, 1:min(6, ncol(abundance_data))])
```

```
##
## 1
## 2                                                                                                d__Archaea
## 3                                                                        d__Archaea|p__Methanobacterio
## 4                                                  d__Archaea|p__Methanobacteriota|c__Methanobacteria|
## 5                          d__Archaea|p__Methanobacteriota|c__Methanobacteria|o__Methanobacteriales|f
## 6 d__Archaea|p__Methanobacteriota|c__Methanobacteria|o__Methanobacteriales|f__Methanobacteriaceae|g_
##   SRR30598618 SRR30598619 SRR30598620 SRR30598621 SRR30598622
## 1           0           0       0.022           0           0
## 2           0           0       0.022           0           0
## 3           0           0       0.022           0           0
## 4           0           0       0.022           0           0
## 5           0           0       0.022           0           0
## 6           0           0       0.000           0           0
```

```r
head(metadata)
```

```
##                   sample sample_name country studyid uric_se1 group acidotic
## SRR30598656 CM_1_met_abs    CM_1_S71  Uganda    1109      1.8     1        0
## SRR30598652 CM_2_met_abs    CM_2_S72  Uganda    1199      3.7     1        0
## SRR30598651 CM_3_met_abs    CM_3_S73  Uganda    1384      6.8     1        0
## SRR30598648 CM_4_met_abs    CM_4_S74  Uganda    1389      8.5     1        0
## SRR30598640 CM_5_met_abs    CM_5_S75  Uganda    1450      3.0     1        0
```

```
## SRR30598628 CM_6_met_abs    CM_6_S76  Uganda    1600      5.7     1        0
##             ceftri ceftri1 ceftri2 antibiot2 antibiot1 antibio2 chloram cipro
## SRR30598656      0       1       0         0         1        0       0     0
## SRR30598652      0       1       0         0         1        0       0     0
## SRR30598651      0       1       0         0         1        0       0     0
## SRR30598648      0       1       0         1         1        0       0     0
## SRR30598640      0       1       0         1         1        0       0     0
## SRR30598628      0       1       0         0         1        0       0     0
##             xpen gentam flagyl cotrimoxazole        doe  stool_dt0 chloram1
## SRR30598656    0      0      0             0  3/24/2015  3/28/2015        0
## SRR30598652    0      0      0             0  3/22/2016  3/24/2016        0
## SRR30598651    0      0      0             0 03/03/2016 03/07/2016        0
## SRR30598648    0      0      0             0  2/24/2016  2/25/2016        0
## SRR30598640    0      0      0             0  9/17/2016  9/21/2016        0
## SRR30598628    0      0      0             0  1/26/2017  1/26/2017        0
##             cipro1 xpen1 gentam1 flagyl1 amoxyl lacidosis tff3_pl1 outcome
## SRR30598656      0     0       0       0      0         0    1.259       0
## SRR30598652      0     0       1       0      0         0    1.774       0
## SRR30598651      0     0       0       0      0         0    2.184       0
## SRR30598648      0     0       0       0      0         0    1.849       0
## SRR30598640      0     0       0       0      0         0    1.674       0
## SRR30598628      0     0       0       0      0         0    2.143       0
##             sm_death12m site sex  age hbsgt haki inpatmal12m inpatmalno12m
## SRR30598656           0    1   1 3.46     1    0           0             0
## SRR30598652           0    2   1 3.97     1    0           1             1
## SRR30598651           0    1   1 3.67     1    0           1             1
## SRR30598648           0    1   2 3.86     1    0           0             0
## SRR30598640           0    1   1 3.87     1    0           0             0
## SRR30598628           0    1   2 3.09     1    0          NA            NA
##             inpatany12m enr_sma hgbbase feedhour neuabs  ho1_pl1 blcx_pos   wbc
## SRR30598656           0       0     8.5       48   4.26  61.5300        0  6.93
## SRR30598652           1       0     9.6       53   4.17  58.8200        0  5.84
## SRR30598651           1       0     7.4       32   5.57 113.8275        0  6.79
## SRR30598648           0       0     7.6       24   9.61  73.6125        0 15.87
## SRR30598640           0       0     7.3       18   8.46 102.0850        0 13.22
## SRR30598628          NA       0     7.3       24   3.24   7.9700        0  6.26
##                     dod trem_pl1 inpatsepsis12m deathdaysto12m blcx_result
## SRR30598656  3/30/2015   278.35              0             NA          NA
## SRR30598652 04/05/2016   425.56              0             NA          NA
## SRR30598651 03/09/2016   356.35              0             NA          NA
## SRR30598648  2/28/2016   265.79              0             NA           6
## SRR30598640  9/22/2016   197.67              0             NA          NA
## SRR30598628  1/31/2017   203.40             NA             NA          NA
##             blcx_result_spec othantibi  whz0  haz0  waz0 pcr_fal0   doe_date
## SRR30598656               NA        NA -0.78 -2.54 -1.96        1 2015-03-24
## SRR30598652               NA        NA -2.03 -0.29 -1.49        1 2016-03-22
## SRR30598651               NA        NA  0.71 -2.58 -1.03        1 2016-03-03
## SRR30598648               NA        NA -1.18 -1.09 -1.45        1 2016-02-24
## SRR30598640               NA        NA -0.99 -2.91 -2.35        1 2016-09-17
## SRR30598628               NA        NA -1.38 -1.16 -1.59        1 2017-01-26
##             admission_month    Sex   Site hbsgt_bin age_in_months
## SRR30598656               3   Male Mulago     AA/AS            42
## SRR30598652               3   Male  Jinja     AA/AS            48
## SRR30598651               3   Male Mulago     AA/AS            44
```

```
## SRR30598648                2 Female Mulago     AA/AS            46
## SRR30598640                9   Male Mulago     AA/AS            46
## SRR30598628                1 Female Mulago     AA/AS            37
##             prior_abx_oral date_to_stool stool_day sick
## SRR30598656              0             4     day 4   SM
## SRR30598652              0             2     day 2   SM
## SRR30598651              0             4     day 4   SM
## SRR30598648              0             1     day 1   SM
## SRR30598640              0             4     day 4   SM
## SRR30598628              0             0     day 0   SM
```

```r
# Check overlap between abundance table samples and metadata row names
colnames(metadata)
```

```
##  [1] "sample"           "sample_name"      "country"          "studyid"
##  [5] "uric_se1"         "group"            "acidotic"         "ceftri"
##  [9] "ceftri1"          "ceftri2"          "antibiot2"        "antibiot1"
## [13] "antibio2"         "chloram"          "cipro"            "xpen"
## [17] "gentam"           "flagyl"           "cotrimoxazole"    "doe"
## [21] "stool_dt0"        "chloram1"         "cipro1"           "xpen1"
## [25] "gentam1"          "flagyl1"          "amoxyl"           "lacidosis"
## [29] "tff3_pl1"         "outcome"          "sm_death12m"      "site"
## [33] "sex"              "age"              "hbsgt"            "haki"
## [37] "inpatmal12m"      "inpatmalno12m"    "inpatany12m"      "enr_sma"
## [41] "hgbbase"          "feedhour"         "neuabs"           "ho1_pl1"
## [45] "blcx_pos"         "wbc"              "dod"              "trem_pl1"
## [49] "inpatsepsis12m"   "deathdaysto12m"   "blcx_result"      "blcx_result_spec"
## [53] "othantibi"        "whz0"             "haz0"             "waz0"
## [57] "pcr_fal0"         "doe_date"         "admission_month"  "Sex"
## [61] "Site"             "hbsgt_bin"        "age_in_months"    "prior_abx_oral"
## [65] "date_to_stool"    "stool_day"        "sick"
```

```r
rownames(metadata)[1:10]
```

```
##  [1] "SRR30598656" "SRR30598652" "SRR30598651" "SRR30598648" "SRR30598640"
##  [6] "SRR30598628" "SRR30598641" "SRR30598634" "SRR30598643" "SRR30598629"
```

```r
colnames(abundance_data)[1:10]
```

```
##  [1] "clade_name"  "SRR30598618" "SRR30598619" "SRR30598620" "SRR30598621"
##  [6] "SRR30598622" "SRR30598623" "SRR30598624" "SRR30598625" "SRR30598626"
```

```r
common_samples_check <- intersect(colnames(abundance_data)[-1], rownames(metadata))
length(common_samples_check)
```

```
## [1] 37
```

```r
head(common_samples_check)
```

```
## [1] "SRR30598618" "SRR30598619" "SRR30598620" "SRR30598621" "SRR30598622"
## [6] "SRR30598623"
```

```r
# ==========================================
# CREATE A PHYLOSEQ OBJECT
# ==========================================

# First column contains taxonomic labels
otu_tab <- abundance_data %>%
  column_to_rownames("clade_name")

# Convert to matrix
otu_mat <- as.matrix(otu_tab)

# Keep only species-level taxa
otu_mat_species <- otu_mat[grepl("\\|s__[^|]+$", rownames(otu_mat)), , drop = FALSE]

# Align samples between abundance matrix and metadata
common_samples <- intersect(colnames(otu_mat_species), rownames(metadata))
length(common_samples)
```

```
## [1] 37
```

```r
head(common_samples)
```

```
## [1] "SRR30598618" "SRR30598619" "SRR30598620" "SRR30598621" "SRR30598622"
## [6] "SRR30598623"
```

```r
otu_mat_species <- otu_mat_species[, common_samples, drop = FALSE]
metadata_sub <- metadata[common_samples, , drop = FALSE]

# Sanity check
stopifnot(all(colnames(otu_mat_species) == rownames(metadata_sub)))

# Build taxonomy table from clade strings
tax_split <- strsplit(rownames(otu_mat_species), "\\|")
max_len <- max(lengths(tax_split))

tax_mat <- do.call(rbind, lapply(tax_split, function(x) {
  c(x, rep(NA, max_len - length(x)))
}))

colnames(tax_mat) <- c("Kingdom", "Phylum", "Class", "Order", "Family", "Genus", "Species", "Strain")[1
rownames(tax_mat) <- rownames(otu_mat_species)

# Create phyloseq object
ps <- phyloseq(
  otu_table(otu_mat_species, taxa_are_rows = TRUE),
  sample_data(metadata_sub),
  tax_table(as.matrix(tax_mat))
)

# Remove samples with missing disease label
ps <- subset_samples(ps, !is.na(sick))
```

```r
# Ensure disease group is a factor
sample_data(ps)$sick <- factor(sample_data(ps)$sick, levels = c("CC", "SM"))

# Inspect object
ps
```

```
## phyloseq-class experiment-level object
## otu_table()   OTU Table:         [ 1974 taxa and 37 samples ]
## sample_data() Sample Data:       [ 37 samples by 67 sample variables ]
## tax_table()   Taxonomy Table:    [ 1974 taxa by 7 taxonomic ranks ]
```

```r
nsamples(ps)
```

```
## [1] 37
```

```r
ntaxa(ps)
```

```
## [1] 1974
```

```r
sample_variables(ps)
```

```
##  [1] "sample"           "sample_name"      "country"          "studyid"
##  [5] "uric_se1"         "group"            "acidotic"         "ceftri"
##  [9] "ceftri1"          "ceftri2"          "antibiot2"        "antibiot1"
## [13] "antibio2"         "chloram"          "cipro"            "xpen"
## [17] "gentam"           "flagyl"           "cotrimoxazole"    "doe"
## [21] "stool_dt0"        "chloram1"         "cipro1"           "xpen1"
## [25] "gentam1"          "flagyl1"          "amoxyl"           "lacidosis"
## [29] "tff3_pl1"         "outcome"          "sm_death12m"      "site"
## [33] "sex"              "age"              "hbsgt"            "haki"
## [37] "inpatmal12m"      "inpatmalno12m"    "inpatany12m"      "enr_sma"
## [41] "hgbbase"          "feedhour"         "neuabs"           "ho1_pl1"
## [45] "blcx_pos"         "wbc"              "dod"              "trem_pl1"
## [49] "inpatsepsis12m"   "deathdaysto12m"   "blcx_result"      "blcx_result_spec"
## [53] "othantibi"        "whz0"             "haz0"             "waz0"
## [57] "pcr_fal0"         "doe_date"         "admission_month"  "Sex"
## [61] "Site"             "hbsgt_bin"        "age_in_months"    "prior_abx_oral"
## [65] "date_to_stool"    "stool_day"        "sick"
```

```r
sample_sums(ps)[1:min(10, nsamples(ps))]
```

```
## SRR30598618 SRR30598619 SRR30598620 SRR30598621 SRR30598622 SRR30598623
##     99.9996     99.9999     99.9992    100.0001     99.9997     99.9998
## SRR30598624 SRR30598625 SRR30598626 SRR30598627
##    100.0005    100.0003    100.0005    100.0003
```

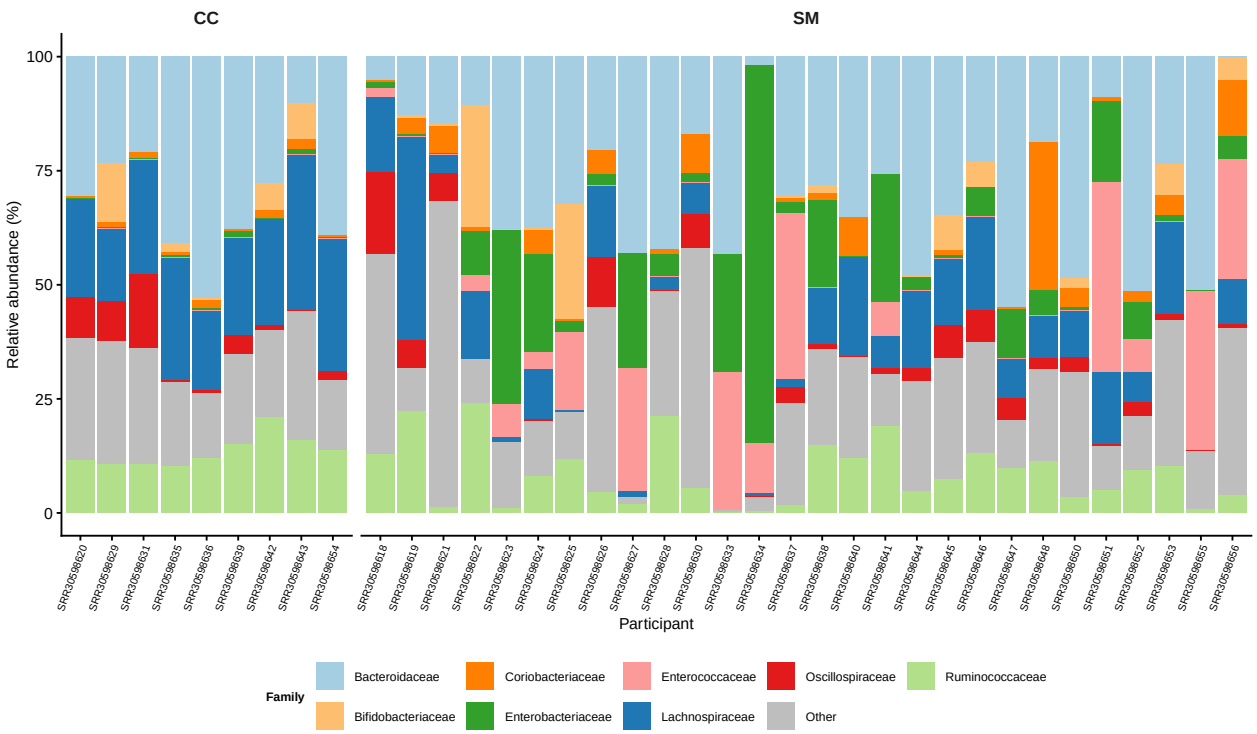
Community composition: sample-level summary

```r
# ============================================
# COMMUNITY COMPOSITION (SAMPLE-LEVEL SUMMARY)
# ============================================

# Clean Family names
tax_table(ps)[, "Family"] <- sub("^f__", "", tax_table(ps)[, "Family"])

# Melt phyloseq object
df <- psmelt(ps)
df$Family <- sub("^f__", "", df$Family)

# Identify top 8 most abundant families
top_families <- df %>%
  filter(!is.na(Family), Family != "") %>%
  group_by(Family) %>%
  summarise(total_abundance = sum(Abundance), .groups = "drop") %>%
  arrange(desc(total_abundance)) %>%
  slice_head(n = 8) %>%
  pull(Family)

top_families
```

```
## [1] "Bacteroidaceae"     "Lachnospiraceae"    "Ruminococcaceae"
## [4] "Enterobacteriaceae" "Enterococcaceae"    "Oscillospiraceae"
## [7] "Bifidobacteriaceae" "Coriobacteriaceae"
```

```r
# Collapse everything else into "Other"
df_plot <- df %>%
  mutate(Family_plot = ifelse(Family %in% top_families, Family, "Other"))

# Order samples by sick group
sample_order <- sample_data(ps) %>%
  data.frame() %>%
  rownames_to_column("Sample") %>%
  arrange(sick) %>%
  pull(Sample)

df_plot$Sample <- factor(df_plot$Sample, levels = sample_order)

# Define colours
families_colors <- c("#a6cee3","#1f78b4","#b2df8a","#33a02c",
                     "#fb9a99","#e31a1c","#fdbf6f","#ff7f00","grey")
families_colors <- setNames(families_colors, c(top_families, "Other"))

# Plot
rel_ab_plot <- ggplot(df_plot, aes(x = Sample, y = Abundance, fill = Family_plot)) +
  geom_bar(stat = "identity", position = "stack") +
  facet_grid(~ sick, scales = "free_x", space = "free_x") +
  scale_fill_manual(values = families_colors, name = "Family") +
  labs(x = "Participant", y = "Relative abundance (%)") +
  theme_classic() +
  theme(
    axis.text.x = element_text(size = 6, angle = 70, hjust = 1, colour = "black"),
```

```
    axis.text.y = element_text(size = 9, colour = "black"),
    axis.title = element_text(size = 9, colour = "black"),
    legend.position = "bottom",
    legend.text = element_text(size = 7),
    legend.title = element_text(size = 7, face = "bold"),
    strip.background = element_blank(),
    strip.text = element_text(size = 10, face = "bold")
  )

rel_ab_plot
```



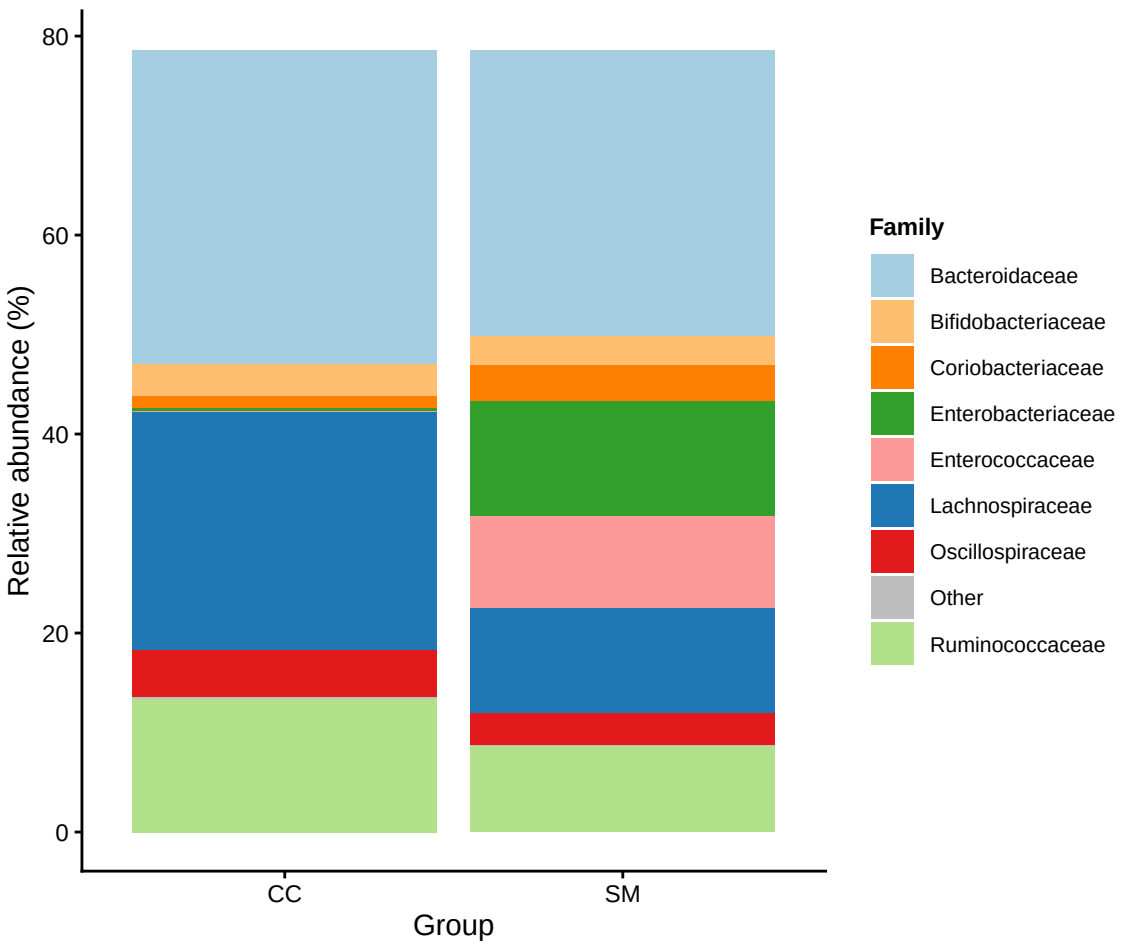

```
ggsave("sample_rel_ab_plot.png", rel_ab_plot, width = 20, height = 13, units = "cm", dpi = 300)
```

Interpretation The gut microbiome varies substantially between children, but a consistent set of dominant families can be seen across samples. Families such as Bacteroidaceae, Lachnospiraceae, Ruminococcaceae, and Enterobacteriaceae contribute substantially to the community structure.

Community composition: group-level summary

```
# ==============================================
# COMMUNITY COMPOSITION (GROUP-LEVEL SUMMARY)
# ==============================================

ps_family <- tax_glom(ps, taxrank = "Family")
df_family <- psmelt(ps_family)

df_family$Family <- sub("^f__", "", df_family$Family)
```

```r
df_family$Family <- ifelse(df_family$Family %in% top_families, df_family$Family, "Other")

df_summary <- df_family %>%
  group_by(sick, Family) %>%
  summarise(mean_abundance = mean(Abundance), .groups = "drop")

rel_ab_group_plot <- ggplot(df_summary, aes(x = sick, y = mean_abundance, fill = Family)) +
  geom_bar(stat = "identity", position = "stack") +
  scale_fill_manual(values = families_colors) +
  labs(x = "Group", y = "Relative abundance (%)") +
  theme_classic() +
  theme(
    legend.position = "right",
    legend.text = element_text(size = 8),
    legend.title = element_text(size = 9, face = "bold")
  )

rel_ab_group_plot
```

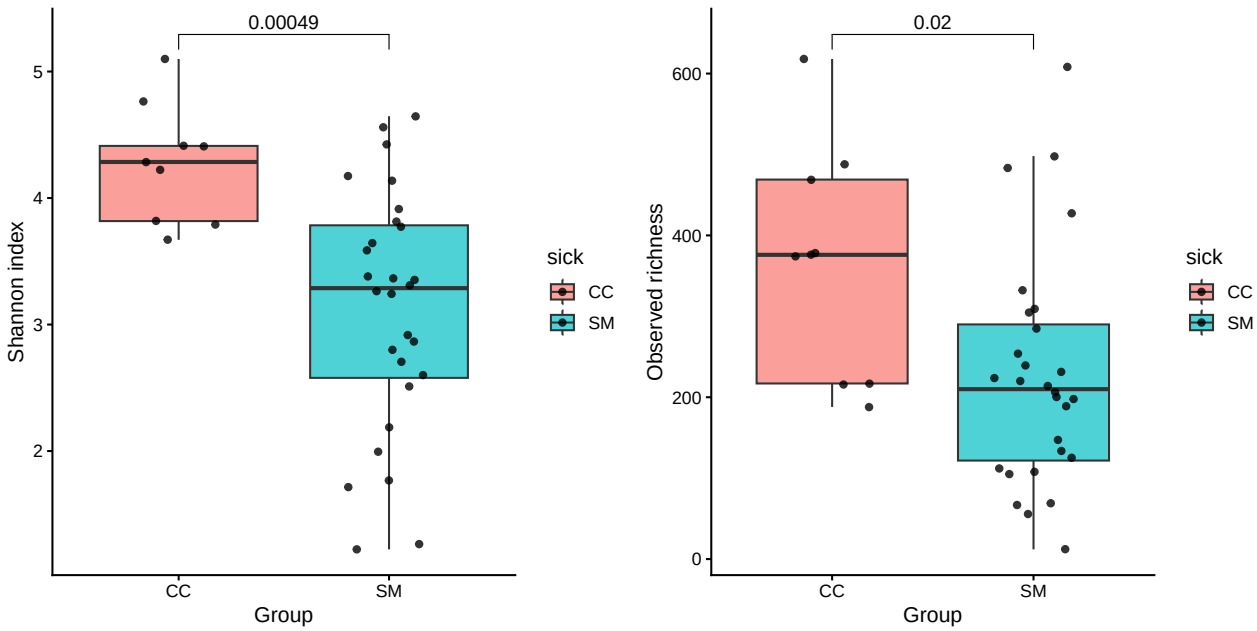



```r
ggsave("group_rel_ab_plot.png", rel_ab_group_plot, width = 8, height = 9, units = "cm", dpi = 300)
```

Interpretation When samples are averaged by clinical group, the relative contribution of major families differs between SM and CC, suggesting broad community restructuring associated with disease status.

Alpha diversity

```r
# ============================================
# ALPHA DIVERSITY
# ============================================

ps_fixed <- ps %>% tax_fix()

alpha_stat_met <- ps_fixed %>%
  ps_calc_diversity(rank = "Species", index = "shannon") %>%
  ps_calc_richness(rank = "Species", index = "observed") %>%
  samdat_tbl()

alpha_stat_met$sick <- factor(alpha_stat_met$sick, levels = c("CC", "SM"))
my_comparisons <- list(c("CC", "SM"))

# Shannon diversity plot
alpha_div_plot_shannon <- ggplot(alpha_stat_met, aes(x = sick, y = shannon_Species, fill = sick)) +
  geom_boxplot(outlier.shape = NA, alpha = 0.7) +
  geom_jitter(width = 0.2, size = 1.5, alpha = 0.8) +
  theme_classic(base_size = 12) +
  labs(x = "Group", y = "Shannon index") +
  scale_fill_manual(values = c("CC" = "#F8766D", "SM" = "#00BFC4")) +
  stat_compare_means(
    comparisons = my_comparisons,
    method = "wilcox.test",
    label = "p.format",
    paired = FALSE
  )

# Observed richness plot
alpha_div_plot_richness <- ggplot(alpha_stat_met, aes(x = sick, y = observed_Species, fill = sick)) +
  geom_boxplot(outlier.shape = NA, alpha = 0.7) +
  geom_jitter(width = 0.2, size = 1.5, alpha = 0.8) +
  theme_classic(base_size = 12) +
  labs(x = "Group", y = "Observed richness") +
  scale_fill_manual(values = c("CC" = "#F8766D", "SM" = "#00BFC4")) +
  stat_compare_means(
    comparisons = my_comparisons,
    method = "wilcox.test",
    label = "p.format",
    paired = FALSE
  )
# Combined figure only
ggpubr::ggarrange(alpha_div_plot_shannon, alpha_div_plot_richness, ncol = 2, nrow = 1)
```




```
ggsave("alpha_diversity_shannon_plot.png", alpha_div_plot_shannon, width = 5, height = 5, dpi = 600)
ggsave("alpha_diversity_richness_plot.png", alpha_div_plot_richness, width = 5, height = 5, dpi = 600)
```

Interpretation Both Shannon diversity and observed richness are lower in SM than in CC, indicating that severe malaria is associated with reduced within-sample diversity and fewer detected taxa.

Beta diversity

```
# ============================================
# BETA DIVERSITY + PERMANOVA
# ============================================

ps_species <- ps %>%
  tax_fix() %>%
  tax_filter(min_prevalence = 1/100, use_counts = FALSE, verbose = FALSE) %>%
  tax_agg(rank = "Species")

ps_clr <- ps_species %>%
  tax_transform("clr", zero_replace = "halfmin")
```

```
## Warning: otu_table of counts is NOT available!
## Available otu_table contains 2711 values that are not non-negative integers
```

```
# Ordination
beta_ordination <- ps_clr %>%
  dist_calc("euclidean") %>%
  ord_calc(method = "PCoA") %>%
  ord_plot(color = "sick", alpha = 0.6, size = 2) +
  theme_bw() +
  coord_fixed(0.7) +
```

```r
  scale_color_manual(values = c("CC" = "#F8766D", "SM" = "#00BFC4")) +
  labs(color = "Group")

# PERMANOVA
perm_vars <- c("sick", "site", "age", "stool_day")
perm_vars <- perm_vars[perm_vars %in% sample_variables(ps_clr)]

perm_res <- ps_clr %>%
  dist_calc("euclidean") %>%
  dist_permanova(
    variables = perm_vars,
    n_perms = 999,
    seed = 123
  ) %>%
  perm_get()
```

```
## 2026-04-13 09:52:57.223618 - Starting PERMANOVA with 999 perms with 1 processes
```

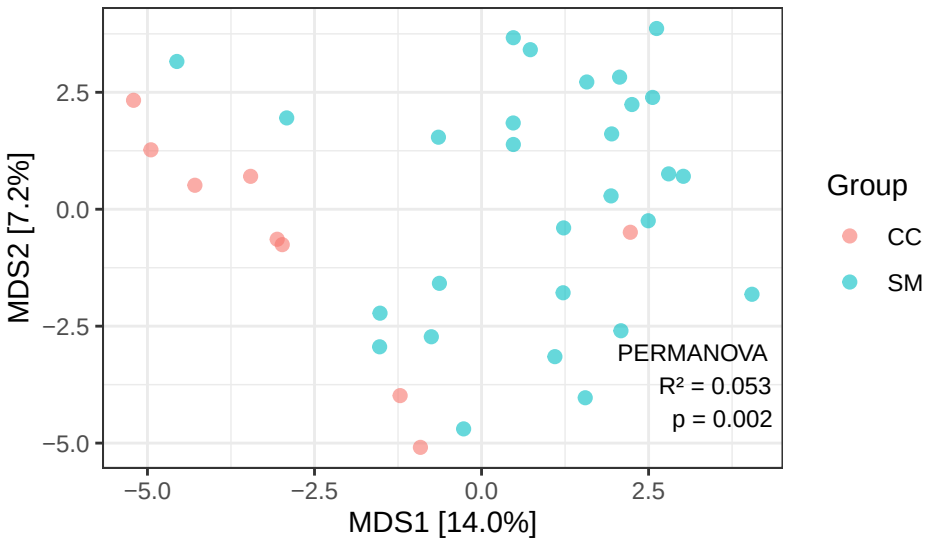
```
## 2026-04-13 09:52:57.534193 - Finished PERMANOVA
```

```r
perm_res
```

```
## Permutation test for adonis under reduced model
## Marginal effects of terms
## Permutation: free
## Number of permutations: 999
##
## vegan::adonis2(formula = formula, data = metadata, permutations = n_perms, by = by, parallel = paral
##           Df SumOfSqs      R2      F Pr(>F)
## sick       1     2730 0.05279 2.0103  0.002 **
## site       1     1460 0.02824 1.0754  0.303
## age        1     1618 0.03129 1.1913  0.144
## stool_day  6     7533 0.14569 0.9246  0.771
## Residual  27    36664 0.70906
## Total     36    51707 1.00000
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# Annotate plot with PERMANOVA result for sick
if ("sick" %in% rownames(perm_res)) {
  pval <- perm_res["sick", "Pr(>F)"]
  r2   <- perm_res["sick", "R2"]

  label_text <- paste0("PERMANOVA\nR² = ", round(r2, 3), "\np = ", signif(pval, 3))

  beta_div_plot <- beta_ordination +
    annotate(
      "text",
      x = Inf, y = -Inf,
      label = label_text,
      hjust = 1.1, vjust = -0.5,
      size = 3
```

```
    ) +
    labs(caption = NULL)
} else {
  beta_div_plot <- beta_ordination
}

beta_div_plot
```




```
ggsave("beta_diversity_plot.png", beta_div_plot, width = 5, height = 5, dpi = 600)
```

Interpretation The PCoA suggests separation between SM and CC in ordination space. PERMANOVA confirms that disease status is significantly associated with overall microbiome composition, although the effect size is modest.

Differential abundance analysis with MaAsLin3

```
# ============================================
# DIFFERENTIAL ABUNDANCE
# ============================================

# Reuse already loaded objects
sylph_data <- abundance_data
study_metadata <- metadata

# Extract taxa and abundance matrix
taxa <- sylph_data[[1]]
abund <- sylph_data[, -1, drop = FALSE]

# Transpose so rows = samples, columns = taxa
abund_t <- as.data.frame(t(abund))
colnames(abund_t) <- taxa

# Check duplicated taxa names
sum(duplicated(taxa))
```

```
## [1] 0
```

```r
# Match samples
common_ids <- intersect(rownames(study_metadata), rownames(abund_t))
length(common_ids)
```

```
## [1] 37
```

```r
head(common_ids)
```

```
## [1] "SRR30598656" "SRR30598652" "SRR30598651" "SRR30598648" "SRR30598640"
## [6] "SRR30598628"
```

```r
metadata_sub <- study_metadata[common_ids, , drop = FALSE]
features <- abund_t[common_ids, , drop = FALSE]

# Check matching order
all(rownames(features) == rownames(metadata_sub))
```

```
## [1] TRUE
```

```r
# Keep only species-level taxa
features_species <- features[, grepl("\\|s__[^|]+$", colnames(features)), drop = FALSE]

# Ensure model variable is a factor
metadata_sub$sick <- factor(metadata_sub$sick, levels = c("CC", "SM"))
```

```r
# Fit MaAsLin3
fit_out <- maaslin3(
  input_data = features_species,
  input_metadata = metadata_sub,
  output = "maaslin3_output",
  fixed_effects = c("sick"),
  normalization = "CLR",
  transform = "NONE",
  min_abundance = 0.001,
  min_prevalence = 0.1,
  warn_prevalence = FALSE,
  cores = 1
)
```

```r
# ============================================
# LOAD AND FILTER RESULTS
# ============================================

res <- read.delim("maaslin3_output/all_results.tsv")

# Keep only sick comparisons
res_sick <- res %>%
  filter(metadata == "sick")
```

```r
# Significant results
sig_res <- res_sick %>%
  filter(qval_individual < 0.25) %>%
  arrange(qval_individual)

# Clean species names
sig_res$species <- sub(".*\\|s__", "", sig_res$feature)

# Inspect
head(sig_res)
```

```
##
## 1                  d__Bacteria|p__Bacteroidota|c__Bacteroidia|o__Bacteroidales|f__Tannerellaceae|g__
## 2               d__Bacteria|p__Pseudomonadota|c__Gammaproteobacteria|o__Enterobacterales|f__Enteroba
## 3                        d__Bacteria|p__Bacillota_A|c__Clostridia|o__Lachnospirales|f__Lachnospirac
## 4                   d__Bacteria|p__Bacillota_A|c__Clostridia|o__Oscillospirales|f__Oscillospiraceae|
## 5 d__Bacteria|p__Bacillota_A|c__Clostridia|o__Lachnospirales|f__Lachnospiraceae|g__Anthropogastromic
## 6                              d__Bacteria|p__Bacteroidota|c__Bacteroidia|o__Bacteroidales|f__Bacter
##   metadata value   name      coef null_hypothesis    stderr pval_individual
## 1     sick    SM sickSM  2.791902      -0.1881342 0.5842996    1.757236e-05
## 2     sick    SM sickSM  2.931664      -0.1881342 0.7133466    1.266848e-04
## 3     sick    SM sickSM -3.617081       0.0000000 1.0344656    4.712826e-04
## 4     sick    SM sickSM -3.617081       0.0000000 1.0344656    4.712826e-04
## 5     sick    SM sickSM -2.126258      -0.1881342 0.5448509    1.884956e-03
## 6     sick    SM sickSM  2.640233      -0.1881342 0.8062366    1.584075e-03
##   qval_individual   pval_joint qval_joint error      model  N N_not_zero
## 1      0.01210735 3.514441e-05 0.01223025  <NA>  abundance 37         34
## 2      0.04364293 2.533536e-04 0.04408353  <NA>  abundance 37         35
## 3      0.08117842 9.423430e-04 0.08198384  <NA> prevalence 37          9
## 4      0.08117842 9.423430e-04 0.08198384  <NA> prevalence 37          9
## 5      0.09990266 3.766359e-03 0.10922440  <NA>  abundance 37         25
## 6      0.09990266 3.165642e-03 0.10922440  <NA>  abundance 37         31
##                              species
## 1          Parabacteroides distasonis
## 2                   Escherichia coli
## 3             Lachnospira sp000436475
## 4           Lawsonibacter sp900549405
## 5 Anthropogastromicrobium sp934292725
## 6                  Bacteroides ovatus
```

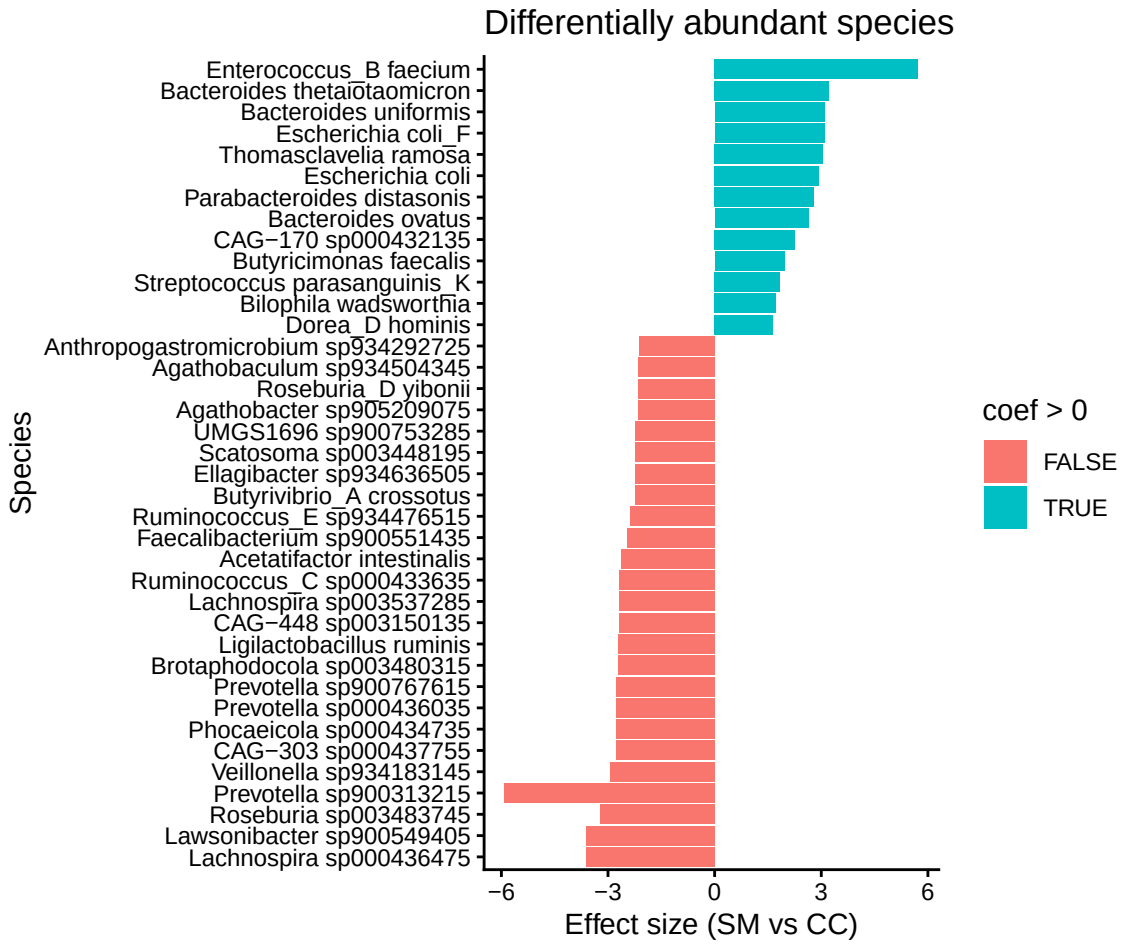
```r
nrow(sig_res)
```

```
## [1] 39
```

Interpretation. MaAsLin3 identifies individual species whose abundance or prevalence is associated with disease status after filtering and compositional normalization. Positive coefficients indicate enrichment in SM, while negative coefficients indicate relative depletion in SM.

Visualise effect sizes (this is what you're missing)

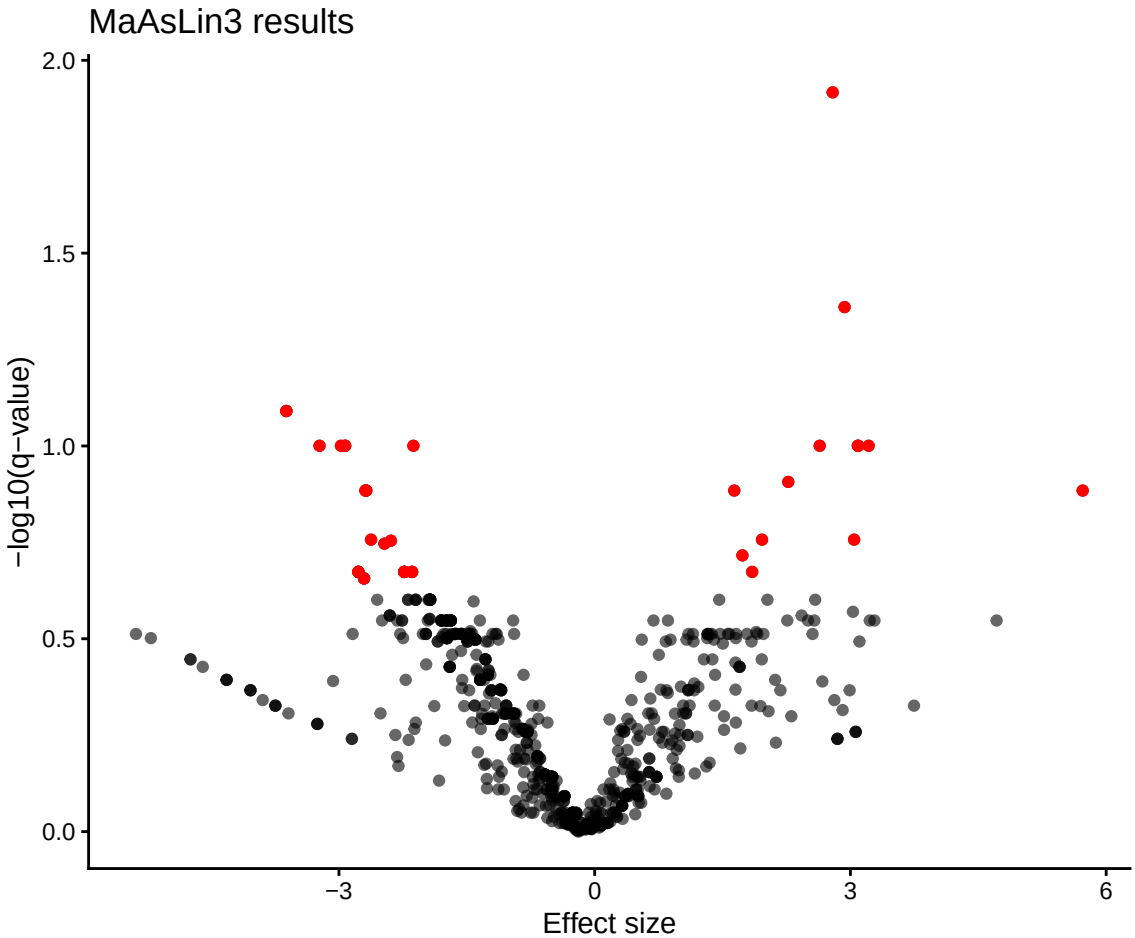
```r
### Option A: simple effect plot
ggplot(sig_res, aes(x = reorder(species, coef), y = coef, fill = coef > 0)) +
  geom_col() +
  coord_flip() +
  scale_fill_manual(values = c("TRUE" = "#00BFC4", "FALSE" = "#F8766D")) +
  labs(
    x = "Species",
    y = "Effect size (SM vs CC)",
    title = "Differentially abundant species"
  ) +
  theme_classic()
```



Interpretation The effect-size plot highlights which significant taxa are enriched or depleted in severe malaria. This helps connect statistical output to biological interpretation.

Option B: volcano-style plot (better for teaching)

```r
res_sick$species <- sub(".*\\|s__", "", res_sick$feature)

ggplot(res_sick, aes(x = coef, y = -log10(qval_individual))) +
  geom_point(alpha = 0.6) +
  geom_point(data = sig_res, color = "red") +
```

```r
  labs(
    x = "Effect size",
    y = "-log10(q-value)",
    title = "MaAsLin3 results"
  ) +
  theme_classic()
```

```
## Warning: Removed 7 rows containing missing values or values outside the scale range
## (`geom_point()`).
```